Activity 5: Doping and Thermal-Budget Interaction

Course: Fundamentals of Semiconductor Device Physics and Process Integration

Course code: TGS-2024049339

Duration: 45 minutes

TSC alignment: A3

PPT alignment: Trainer Slides v7.1, concept slides 210-228 and Activity 5 overview slide 427.

Workplace scenario

Threshold voltage drifts low after a revised rapid thermal anneal recipe is introduced to improve activation.

Goal

Analyse implantation, activation, diffusion and oxide interactions across the thermal budget.

Files in this folder

- README.md / README.pdf - complete procedure and acceptance criteria.
- SCENARIO.md / SCENARIO.pdf - facts, assumptions and role brief.
- WORKSHEET.md / WORKSHEET.pdf - structured learner response.
- EVIDENCE-CHECKLIST.md / EVIDENCE-CHECKLIST.pdf - completion and review record.
- Any CSV file in this folder - supplied teaching data for analysis.

Step-by-step procedure

1. Compare the baseline and revised implant-plus-anneal travellers.
2. List the intended effect and possible side effect of dose, energy, tilt, temperature and time.
3. Calculate the cumulative thermal exposure using the worksheet's relative thermal-budget index.
4. Link each plausible mechanism to sheet resistance, junction depth, threshold voltage or leakage evidence.
5. Propose a split-lot experiment that isolates activation gain from unwanted diffusion.

Required evidence

Thermal-budget comparison, evidence map and split-lot experiment design.

Include your completed worksheet, any calculations or annotated diagrams, the evidence checklist, and a short note identifying assumptions or unresolved evidence gaps.

Acceptance criteria

[ ] Implant and anneal variables are considered together

[ ] At least three electrical or physical measures are selected

[ ] Split-lot factors and response measures are explicit

Verification

Before submission, ask a peer to challenge one causal link. Revise the conclusion if the challenge reveals that evidence and inference have been mixed. Your final response must distinguish: observed fact, interpretation, decision, and follow-up check.

Troubleshooting

- If several mechanisms fit the symptom, choose a measurement that discriminates between them.
- If a process module lacks a defined input, output or control gate, the flow is incomplete.
- If an action is not tied to a verified cause, record it as a hypothesis test rather than corrective action.
- If calculations disagree, show the method and units so the discrepancy can be reviewed.